

Advancements in NDT for Bridges with Hidden Critical Elements; Findings from National Highways Structures Moonshot

THE
**STRUCTURES
MOONSHOT**
PROJECT

BRIDGES
New Civil Engineer
CONFERENCE & AWARDS

Structures Moonshot Data Schedule>

E	F	G	H
Tool	Element	Defect Type	Confidence
MISTRAS Tablet UT	Tendon	Corrosion	High
MISTRAS Tablet UT	Tendon	Good Condition	High
Seisq PD8052		Voids	High
N/A	Concrete	Cracking	High
N/A	Grout	Voids	High

Session Speaker



Chris Mundell

**AtkinsRealis Technical Director, Transportation, UK
Structures Moonshot Project Director**

Scope of the Moonshot Project

PHASE 1 (20-21)

Call for Ideas – Non-Destructive Testing (NDT) Technologies for Post-tensioned Structures

PHASE 2 (23 – 25)

Research & Development on low Technology Readiness Level (TRL) technologies – **guided wave, electrical impedance, and muon tomography**

Evaluating the **efficiency of high TRL NDT Technologies** in detecting hidden defects in structures on the A14 Huntingdon Viaduct Demolished Sections

PHASE 3 (25 – 26)

Deployment of low TRL technologies on **in-service structures** alongside **further R&D**

Technology Investment – Guided Wave

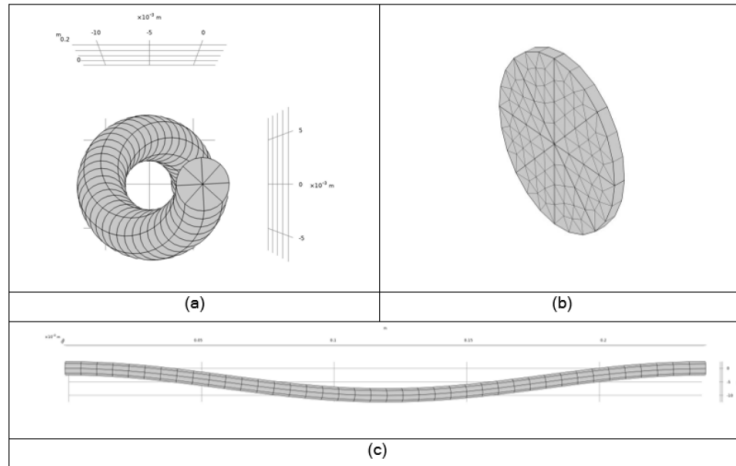
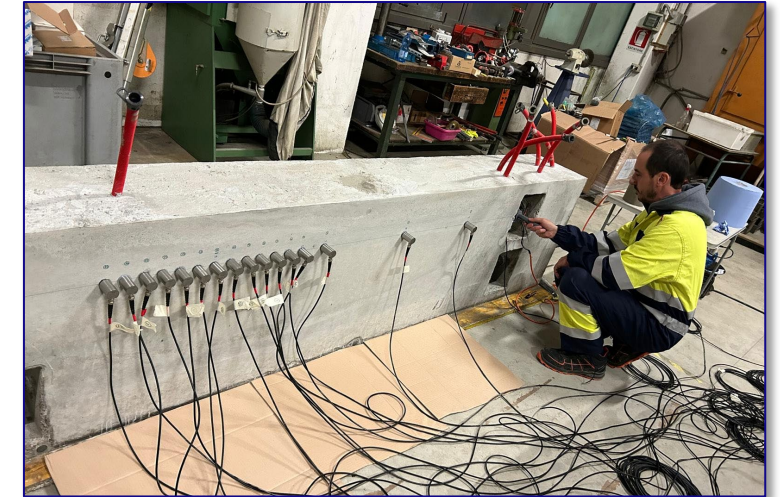
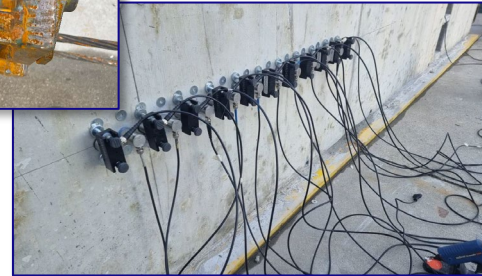


Figure 5. Geometry of the single-wire helical. (a) front view, (b) unit cell used in WFEM method to compute the dispersion curves, (c) side view.



Simulations & Numerical Proof

Physical Technology Development

Test Beam Trials

Investment Milestone # 1

Numerical trials, proof of concept
Technology specs, expectations and
limitations developed

Milestone # 2

Hardware development
Small scale test ratification
Large scale testing planned

Milestone # 3

Large scale test beam
commissioned
Defect identification trials

Technology Investment – Guided Wave



Investment Milestone # 4

Transducer development for real-world applicability
Trials of hardware on test beam

Milestone # 5

Deployment of technology on in-service structure
Controlled PTSI at identified defect locations
Evaluation of technology against performance specification

Global NDT Study– Field Testing

SAMPLE 1



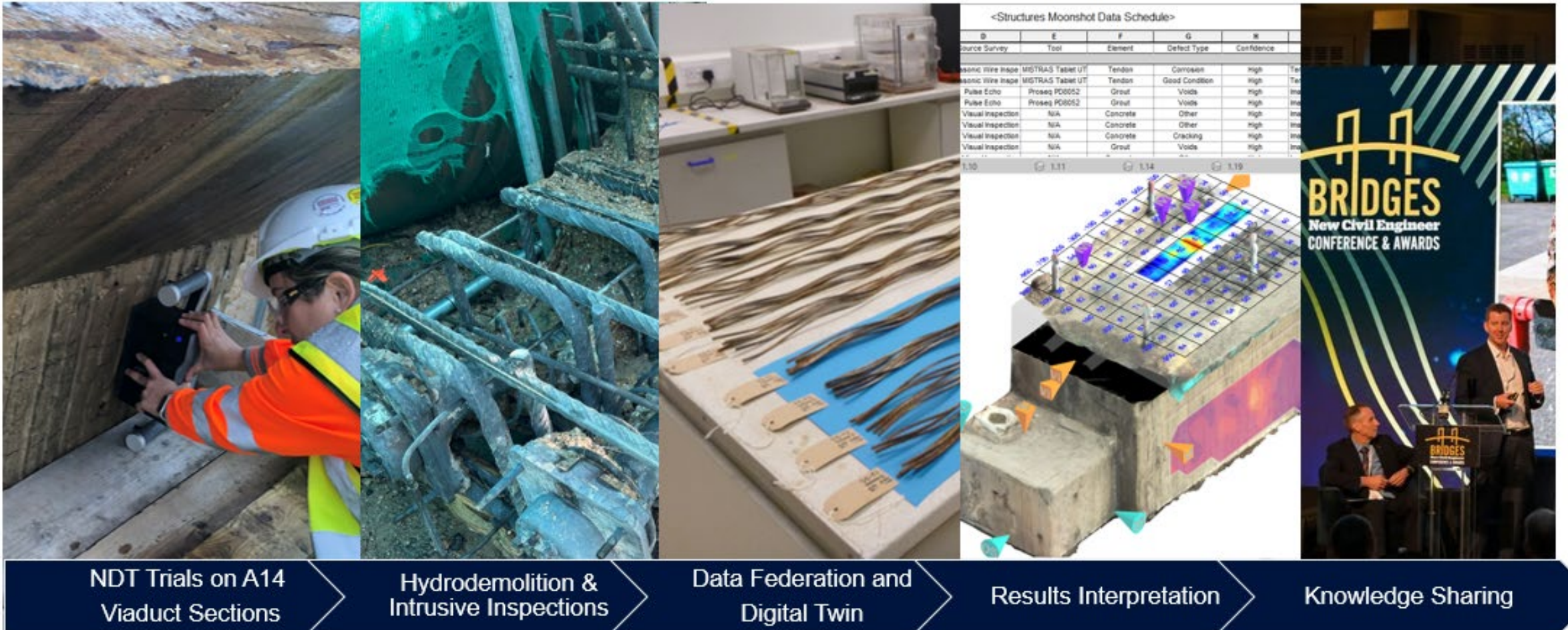
SAMPLE 2



SAMPLE 3



Global NDT Study– Field Testing



Global NDT Study– Field Testing

Hierarchy Level	Description	Applicability		Value
1	Locating and identifying physical properties	HJ	PT	Increasing Value of Data
1.1	Locate the strands and/or the rebar approximately	✓	✓	
1.2	Identify sizes of strand and/or rebar	✓	✓	
1.3	Identify duct type		✓	
1.5	Map all strands and rebar completely and accurately – position and sizes (stretch target)	✓	✓	
1.6	Identify chemical and physical properties of the materials (e.g. Section 3.7.5 and 3.7.6 of CS 465)	✓	✓	
2	Conditions for corrosion			
2.1	Locate any absence of duct grouting		✓	
2.2	Identify areas where the reinforcement is at high risk of corrosion, e.g. loss of concrete passivation or high chlorides	✓	✓	
2.3	Identify areas where the tendons are at high risk of corrosion, e.g. loss of concrete passivation or high chlorides		✓	
3	Indications of damage			
3.1	Identify discontinuities in reinforcing and/or prestressing steel that may be indicative of corrosion or breakages	✓	✓	
3.2	Identification of plastic deformations of reinforcement (e.g. necking)	✓		
3.3	Measurement of surface cracking at re-entrant corners (width, length, depth)	✓		
3.4	Measurement of cracking along length of re-entrant corners (width, length, depth)	✓		
4	Pinpoint location and quantify magnitude of deterioration			
4.1	Corrosion locations	✓	✓	
4.2	Magnitude and shape of section loss from corrosion at each location	✓	✓	
4.3	Wire break locations		✓	
4.4	Number of wire breaks at each location		✓	
5	Other behaviour to detect if possible			
5.1	Bond slip between steel and concrete (should have occurred at least in cutting samples)		✓	
5.2	Re-anchoring of strands (should have occurred at least in cutting samples)		✓	

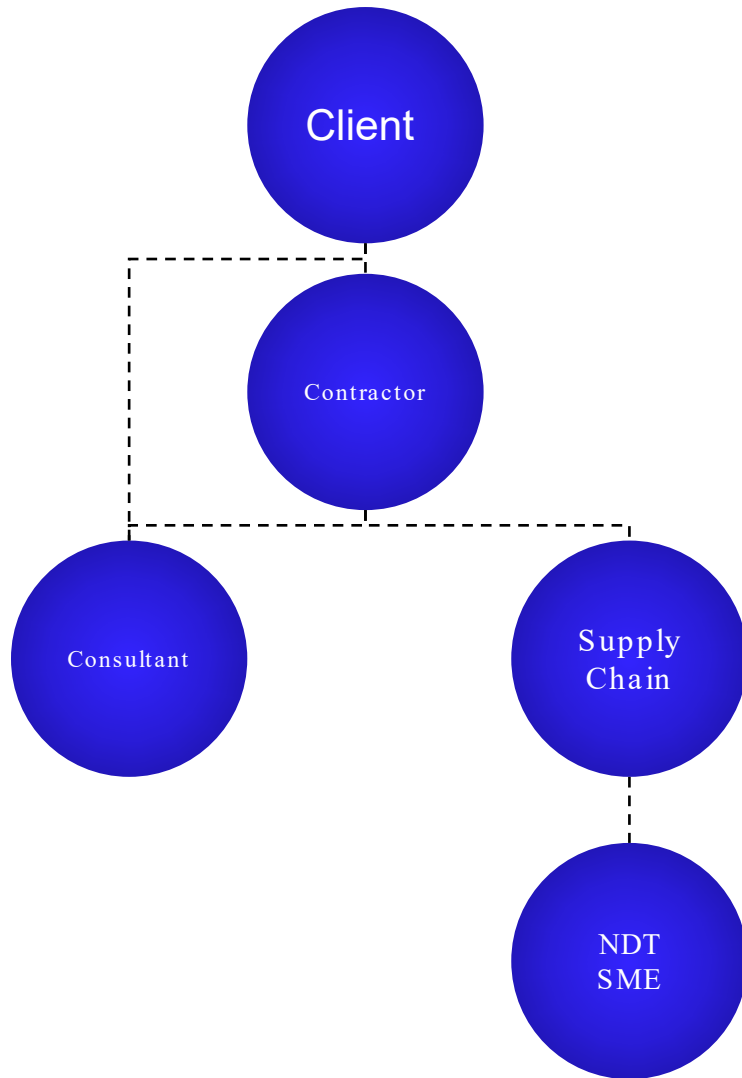
Type of test	Activity/Test Type
Mechanical Wave	Acoustic Emission Monitoring (AEM)
Optical	Initial visual inspection
Electromagnetic	Covermeter survey
Optical	Hyperspectral Imagery
Particle sensors	FTIR hydrogen gas monitoring
Optical	Portable X-Ray (PTX)
Electromagnetic	Chloride mapping
Mechanical Wave	Anchorage guided wave
Mechanical Wave	Advanced Impact Echo
Electromagnetic	Ground Penetrating Radar (GPR)
Optical	Infra-red thermography (IRT)
Electrochemical	Half Cell Potential Survey
Magnetic	Magnetic Flux Leakage (MFL)
Mechanical Wave	Joint fracture evaluation
Mechanical Wave	Advanced ultrasonics (ELOP & Pundit)
Electromagnetic	GSSI StructureScan Mini (GPR)
Core sampling	Chloride core samples
Electrochemical	Electrical resistivity
Optical	DIC (during demolition)
Optical	Demolition & inspection

Project Collaboration

THE
STRUCTURES
MOONSHOT
PROJECT



Project Collaboration



Typical Arrangement

Client specify requirements (potentially including testing requirements), standards to be applied, activities to be undertaken

Consultant will undertake inspections & create testing specification

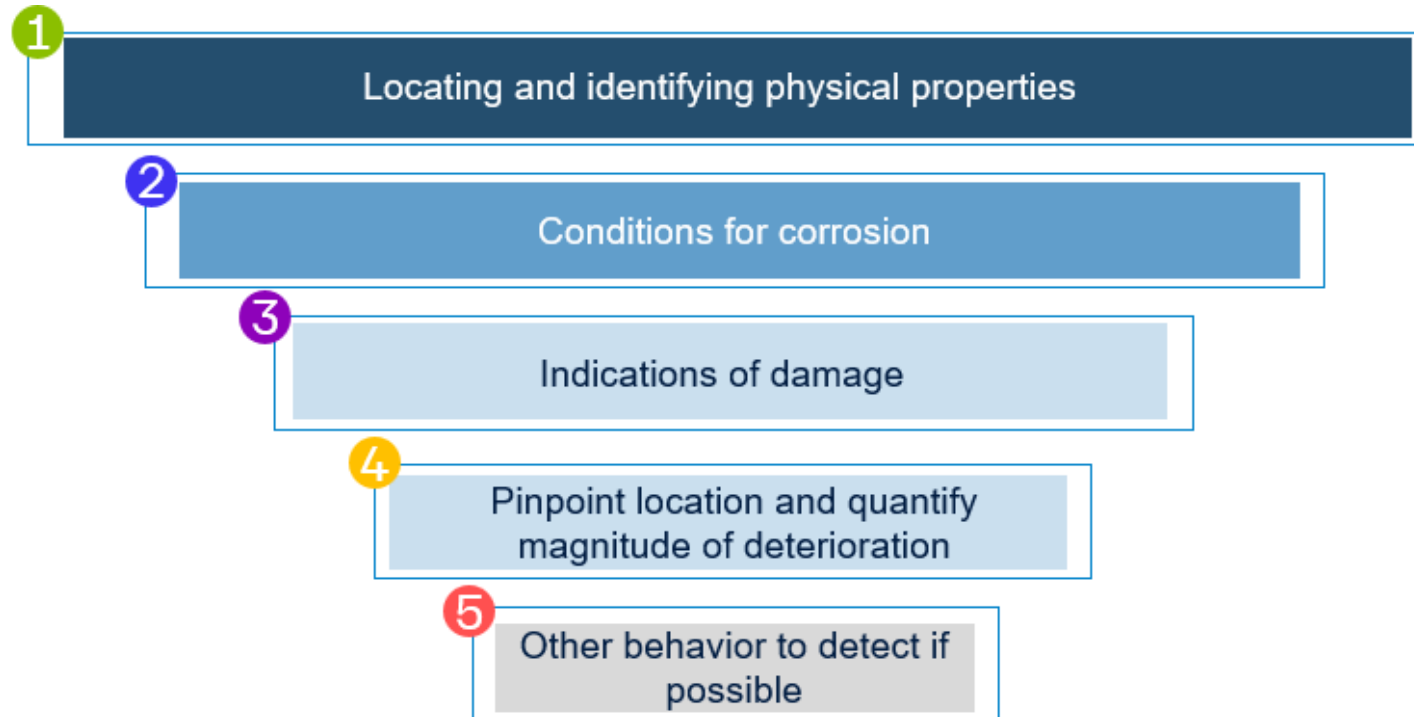
NDT specialist at end of supply chain, often engaged once all requirements have been agreed and with set scope and agreed costs

Global NDT Study– Outcomes

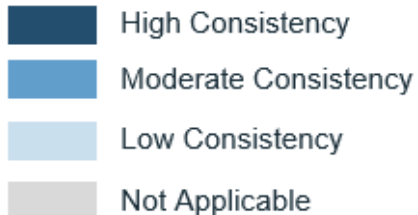
PHASE 2
(23 – 25)

Research & Development on low Technology Readiness Level (TRL) technologies – **guided wave, electrical impedance, and muon tomography**

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More Details



Outcomes - Reporting



Outcomes – Practitioners Toolkit

NDT Practitioners Toolkit
ATKINSREALIS · NATIONAL HIGHWAYS

[Home](#)[Decision Trees](#)[NDT Matrix](#)[Technologies](#)[About](#)[Key](#)[Inspection Regime](#)

Testing Practitioners Toolkit

A practical decision-support resource enabling bridge owners and practitioners to select the right NDT method for post-tensioned concrete bridge assessment.

[Start with Decision Tree →](#)[About this Toolkit](#)



THREE TOOLS, ONE TOOLKIT

1



DECISION TREES

Identify your structural concern and follow the interactive flowchart to the right investigation pathway. Clickable nodes link directly to the NDT Matrix.

8 condition pathways

Open →

2



NDT MATRIX

Match investigation objectives to the most suitable NDT methods. Click any technology to view its full profile and deployment guidance.

5 objective categories

Open →

3



TECHNOLOGY ONE-PAGERS

Detailed profiles for each NDT technology — overview, strengths, limitations, practical tips, data output examples and photographic evidence.

15 technologies profiled

Open →

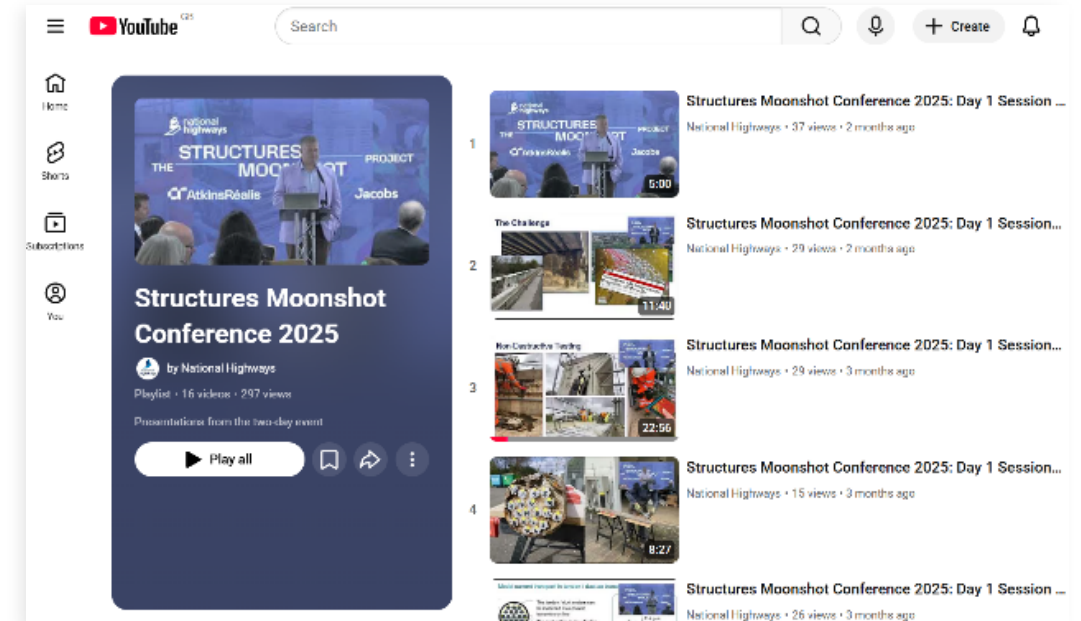
[About & Introduction](#)[Toolkit Key](#)[How to Use](#)

Ref: SPATS2-TO321_AJIV_SBR_XX_DO_CB-0004 · May 2026

Outcomes - Knowledge Sharing

200 Attendees over 2 days

- 16 Client organisations including National Highways, Transport Scotland, Local Authorities, Network Rail, BEAR Scotland, TfL, US DoT & overseas clients
- 11 Contractor organisations including Balfours, Costain, Kier, VSL, Balvac
- 18 Consultants including Aecom, Amey, Arup, Cowi, Mott, WSP, Arcadis
- 11 NDT SMEs including Mistras, Freyssinet, Socotec, Xeiad
- 4 Universities including Idaho, Missouri, Surrey, Southampton
- 6 Exhibitors including Mistras, Hausbots, Hilti, Southampton and Screening Eagle



Phase 3: Delivery on Live Structures

1

Huntingdon Sample
Trials



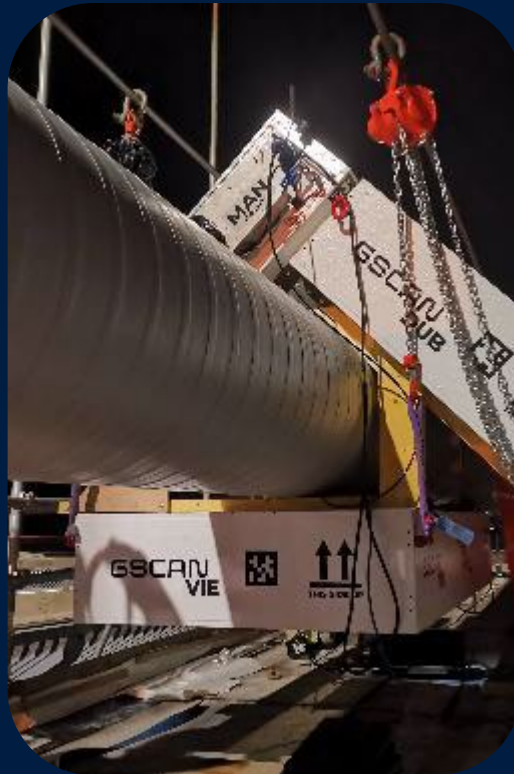
2

Muon Tomography
In-Service Trials



3

Suspension Cable
Trials



4

Materials Testing
Expansion Trials

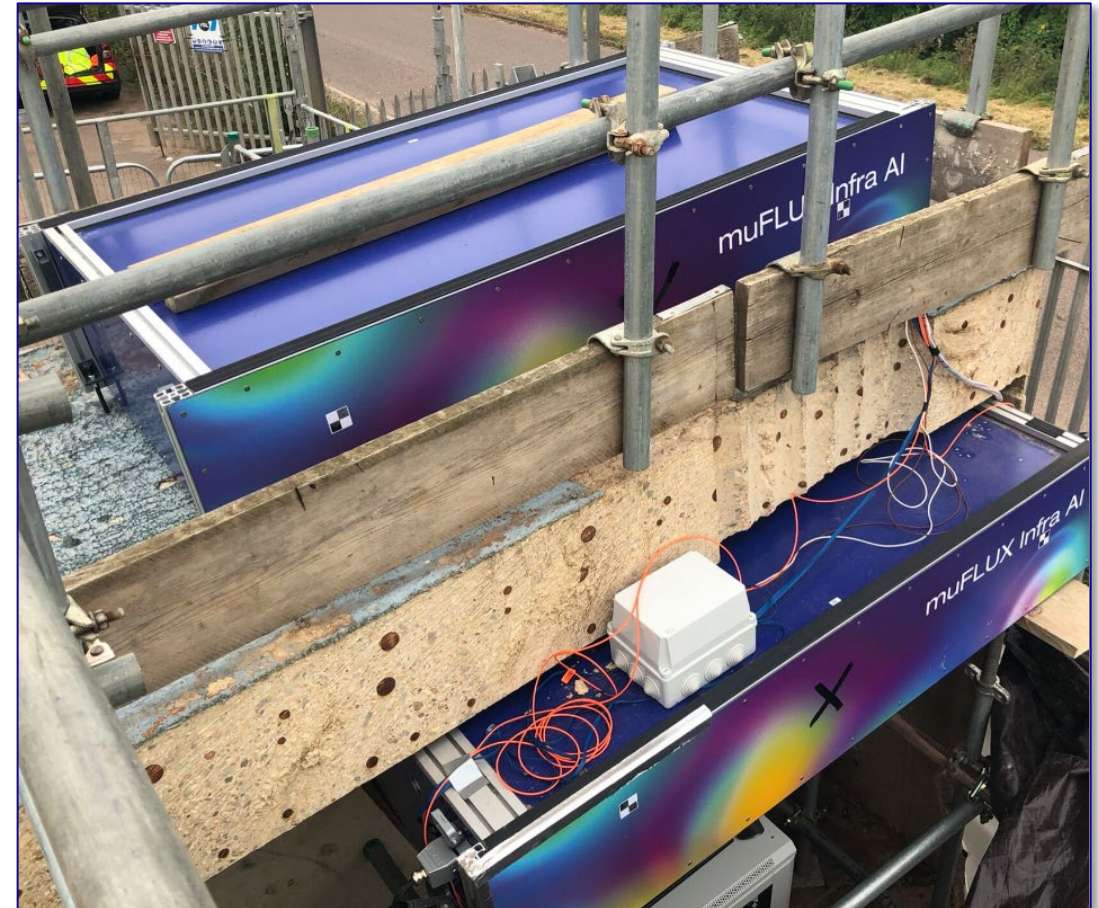
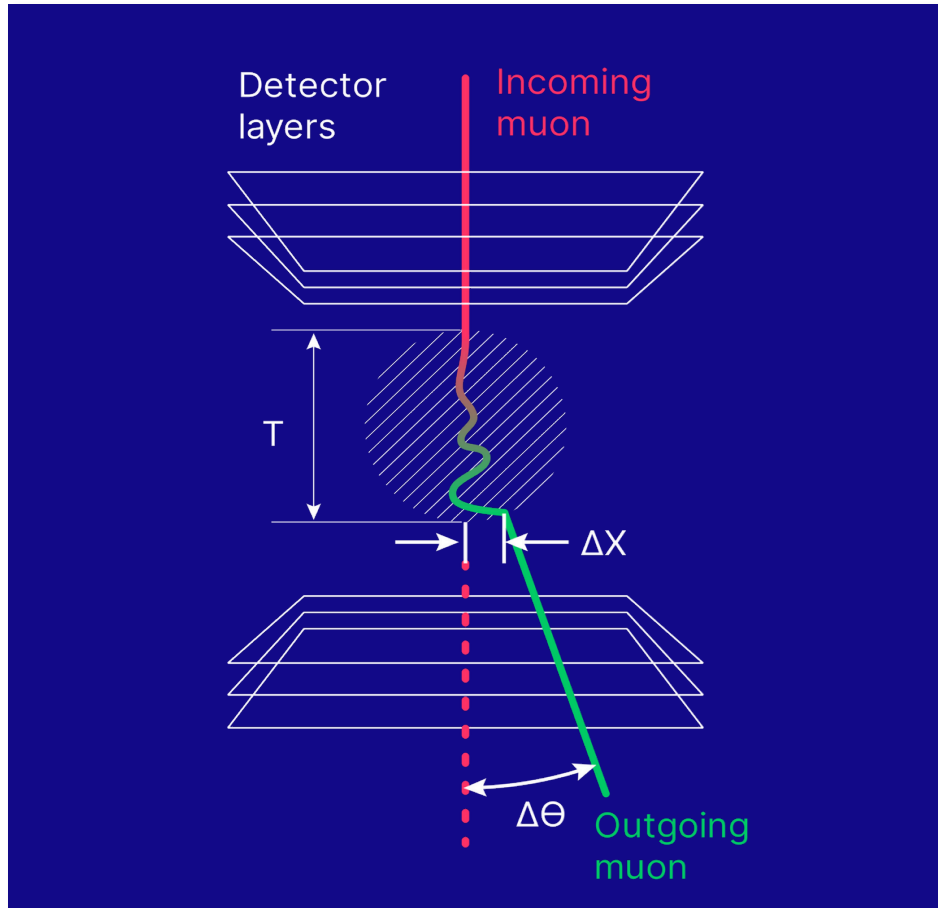


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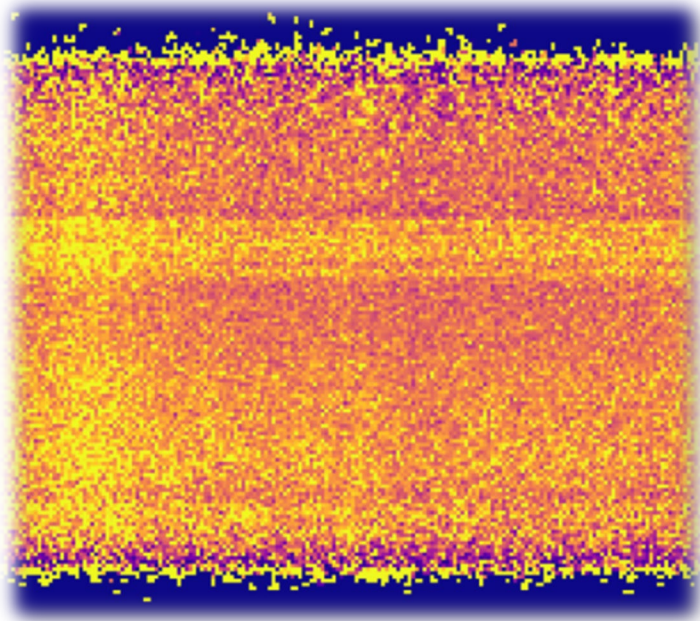
Guided Wave
Trials



Gscan – Technology Overview



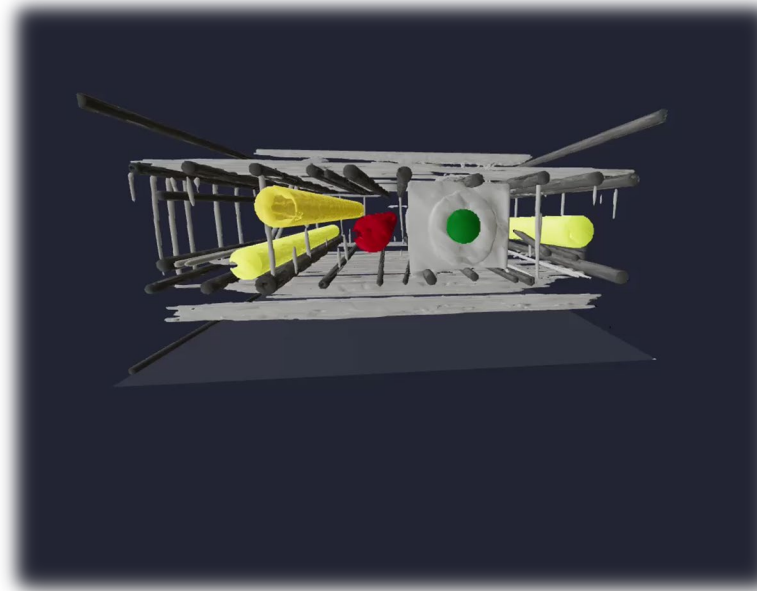
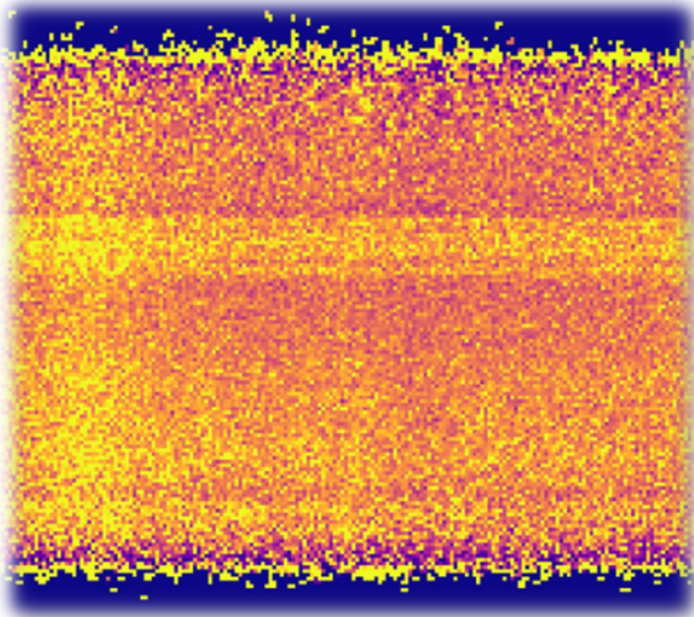
Gscan – Technology Progression



Milestone # 1

Huntingdon Viaduct (HRV) Trials

Gscan – Technology Progression



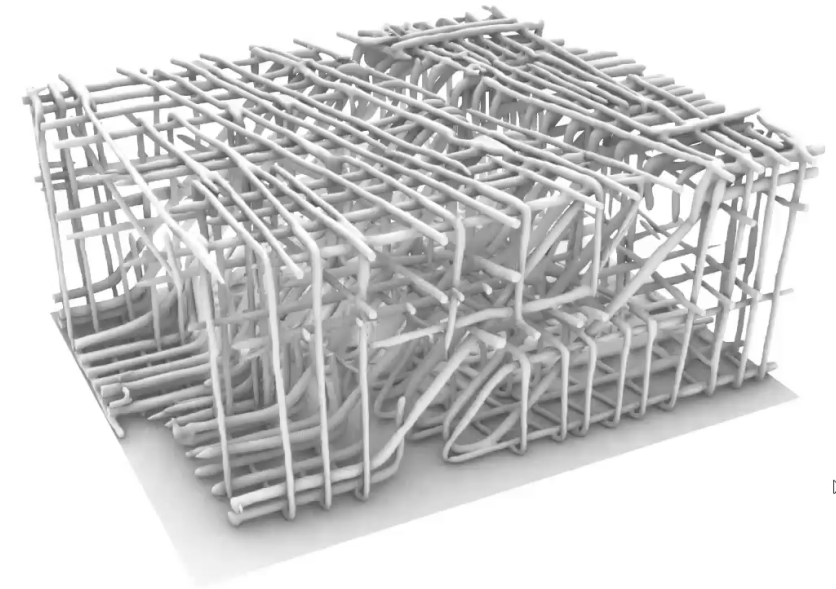
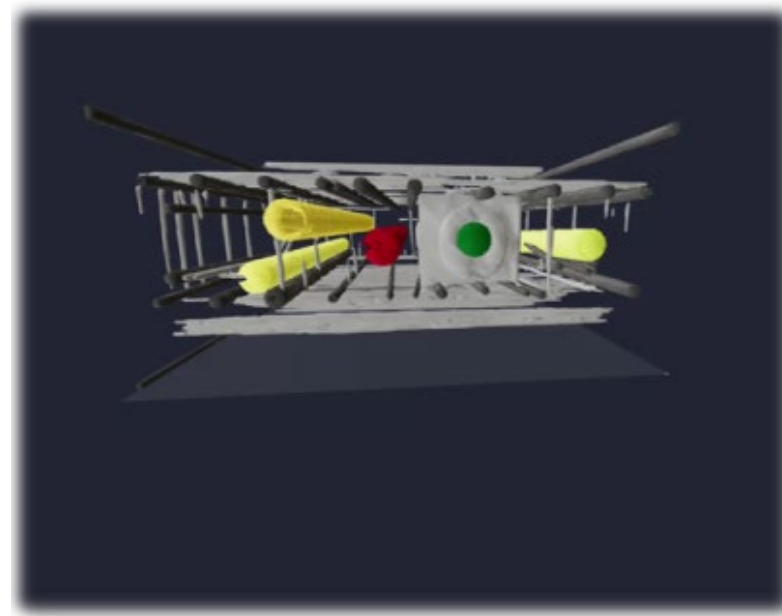
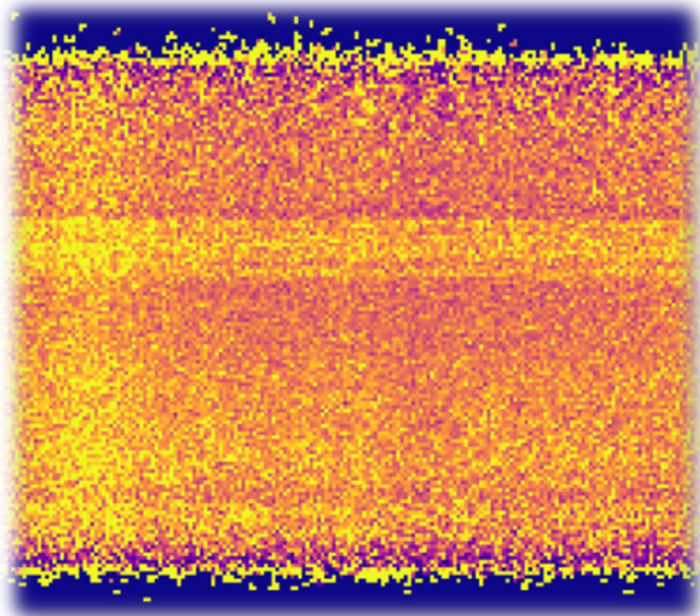
Milestone # 1

Huntingdon Viaduct (HRV) Trials

Milestone # 2

Estonian Test Beam

Gscan – Technology Progression



Milestone # 1

Huntingdon Viaduct (HRV) Trials

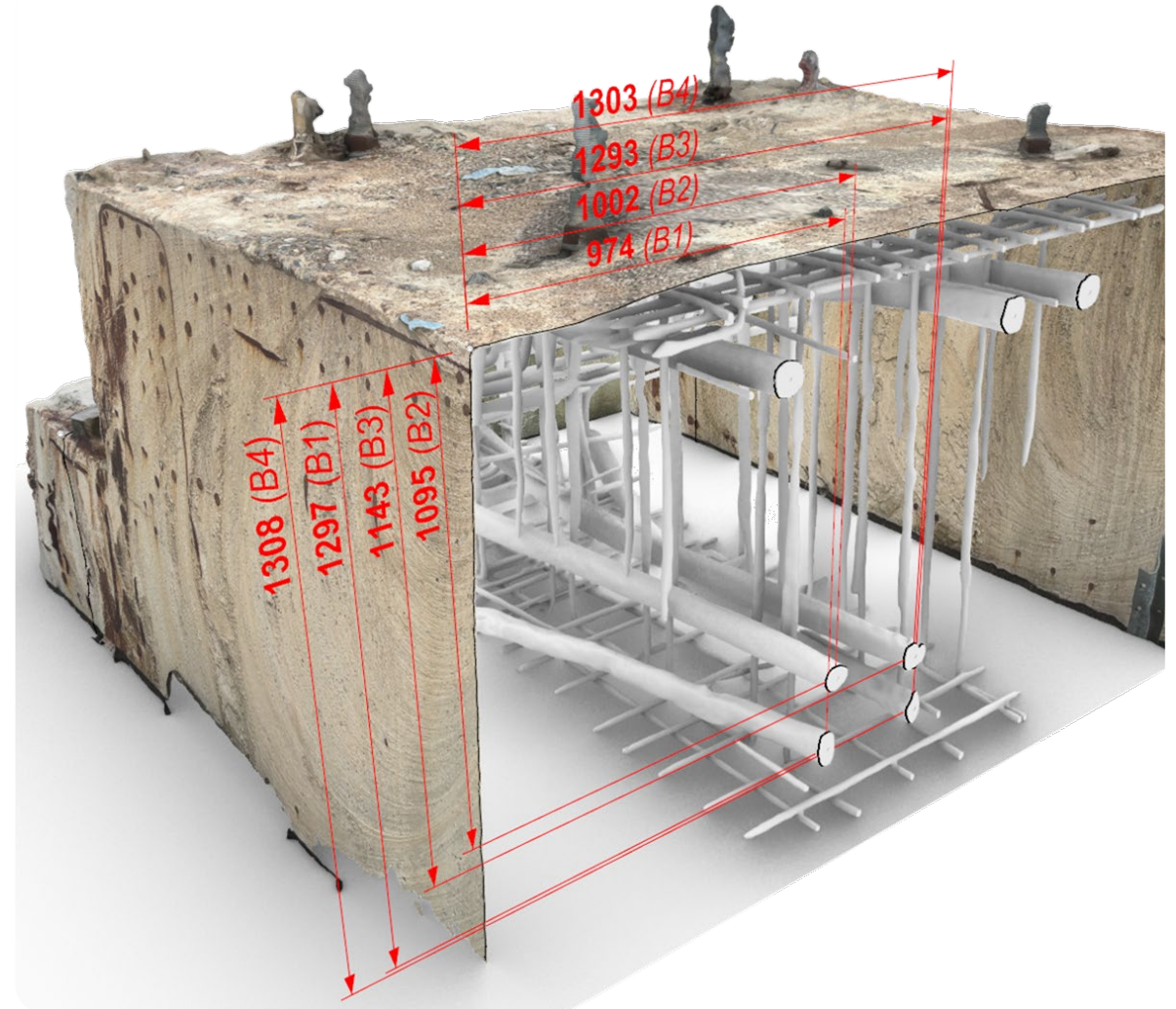
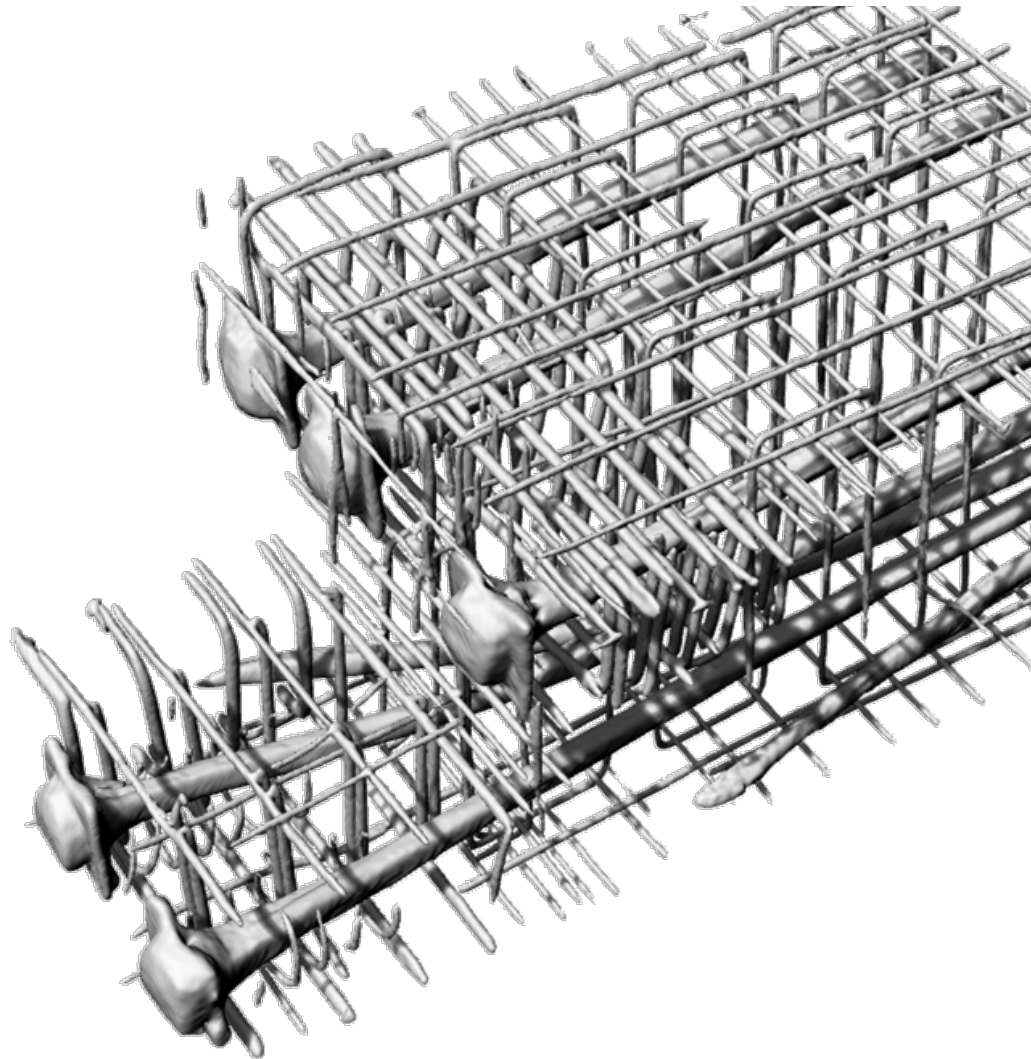
Milestone # 2

Estonian Test Beam

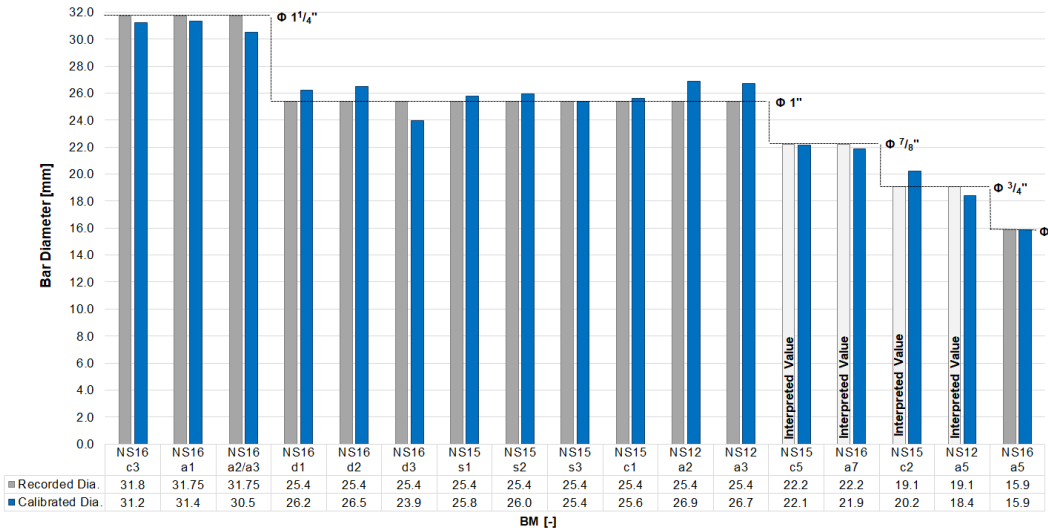
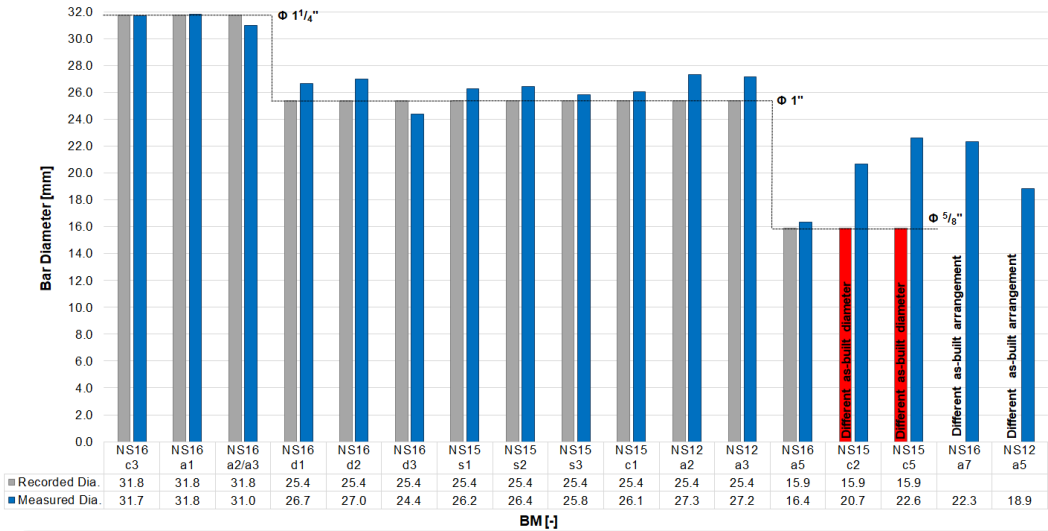
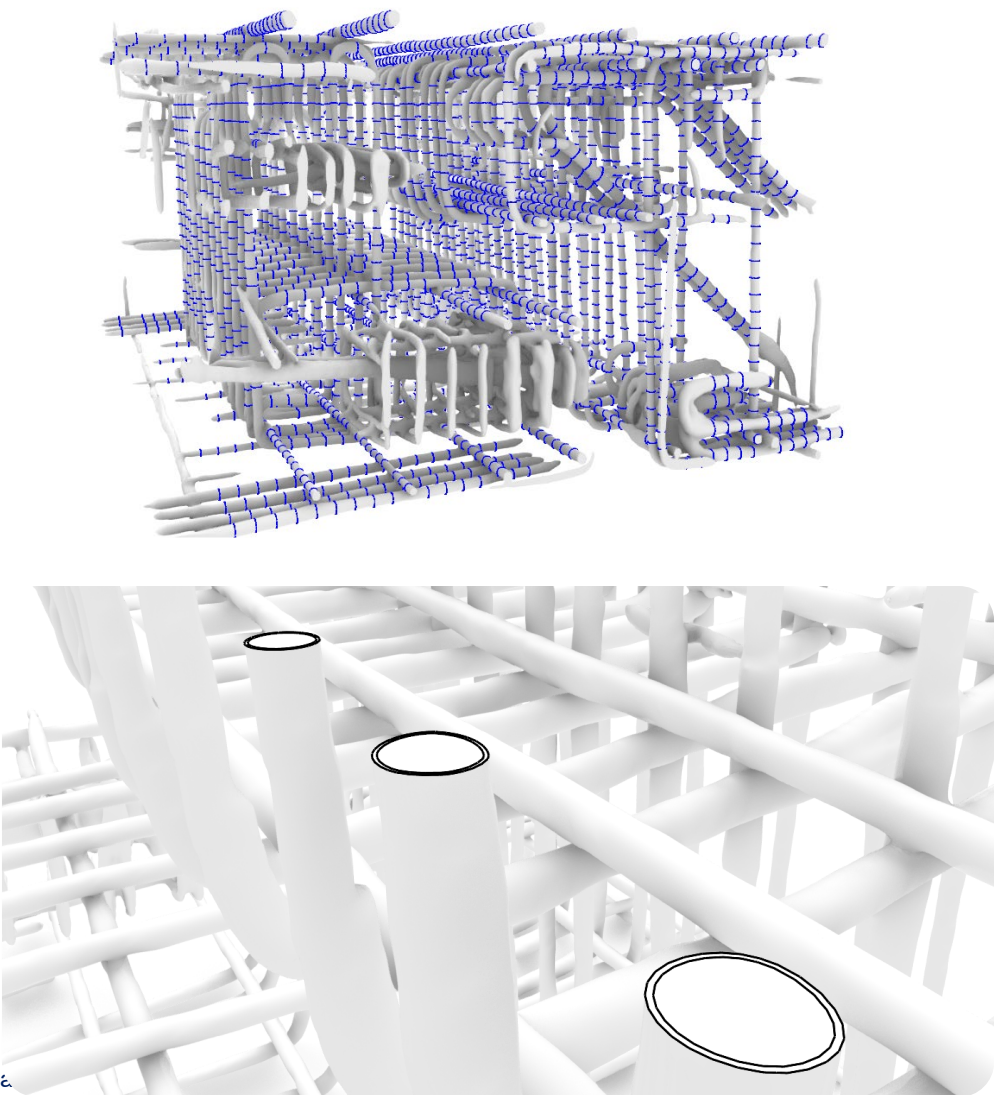
Milestone # 3

Live Trials
Further HRV Trials

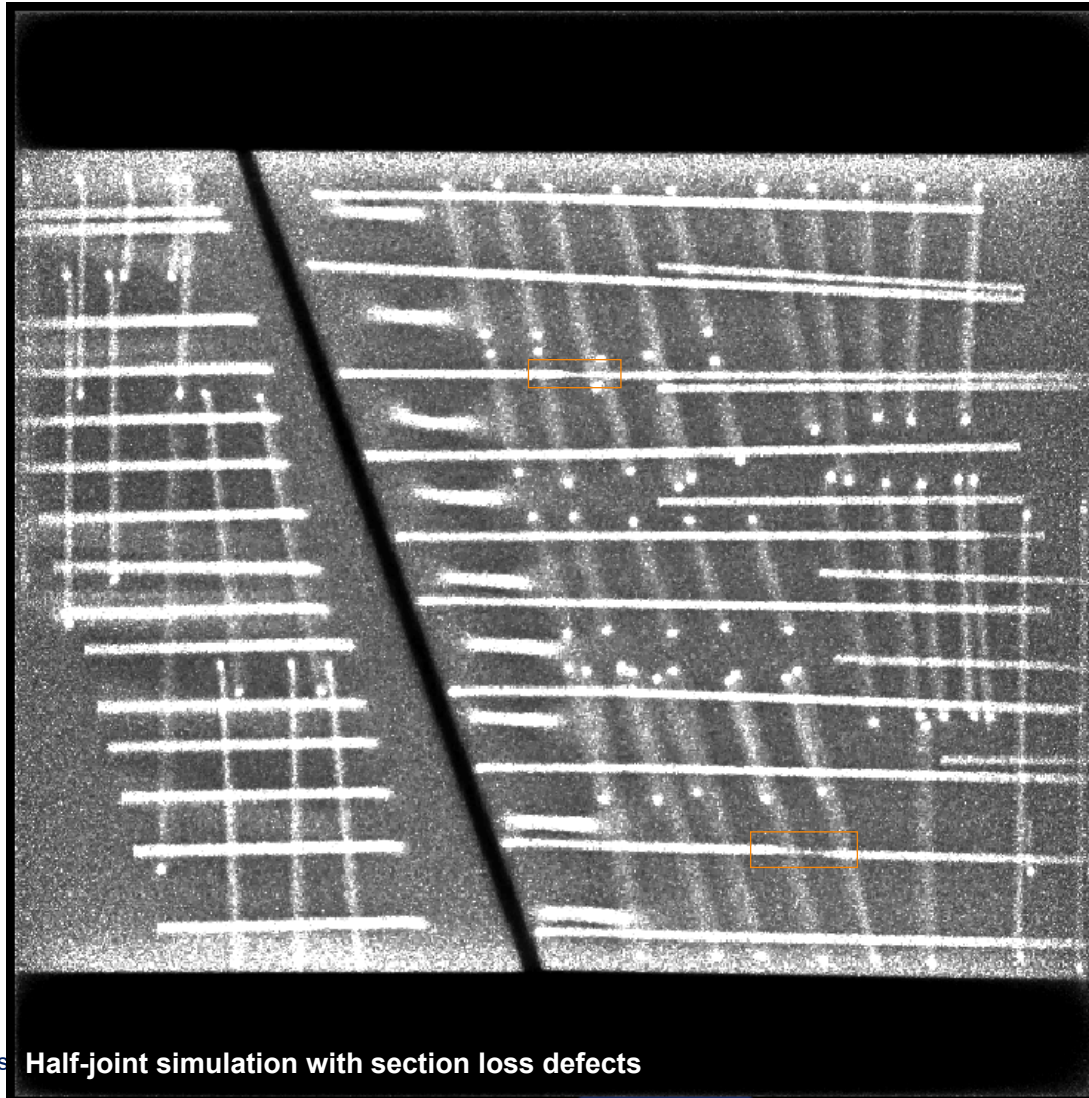
Gscan – Latest Trials on HRV Sample



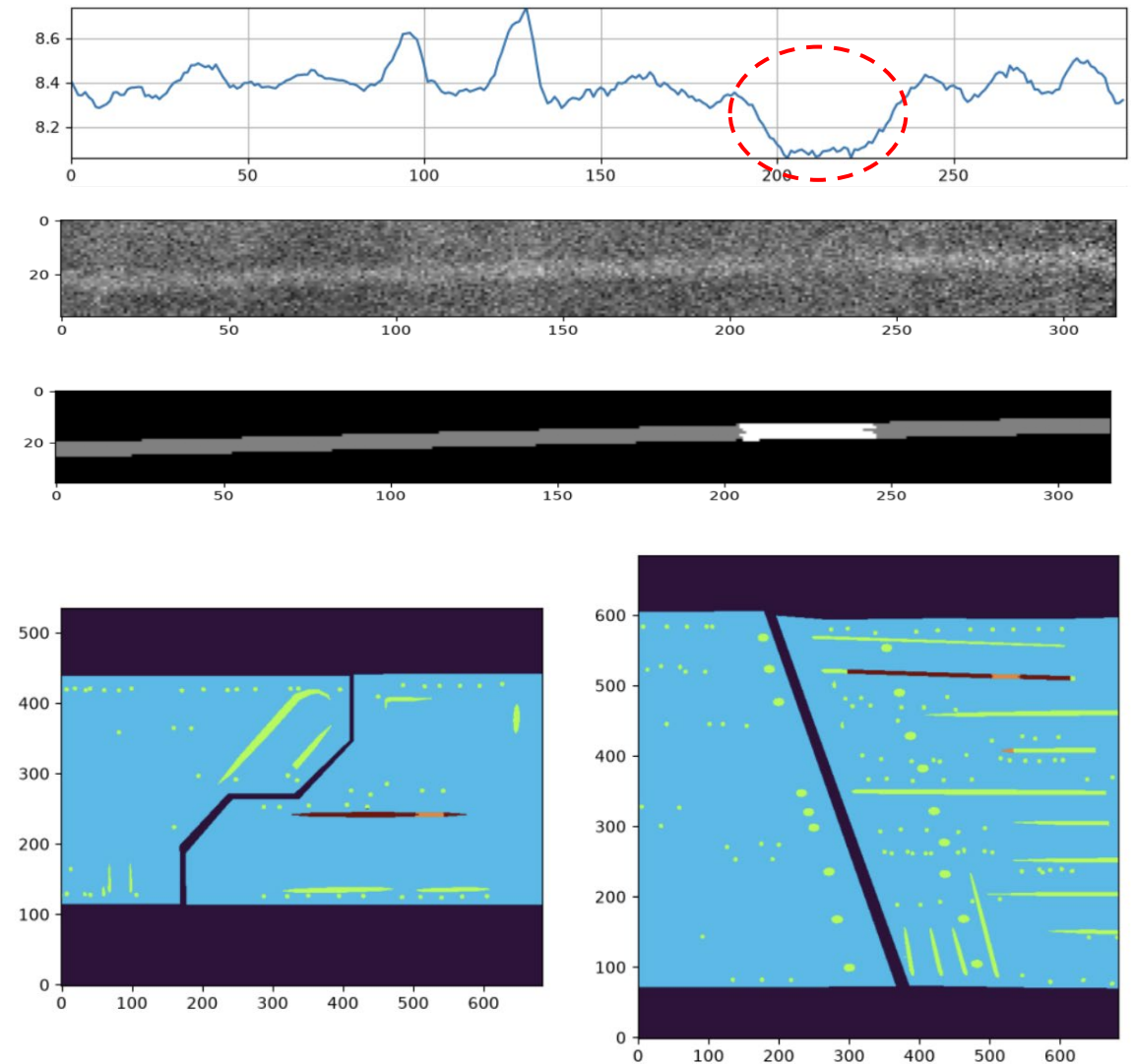
Gscan – Rebar Diameter Calibration



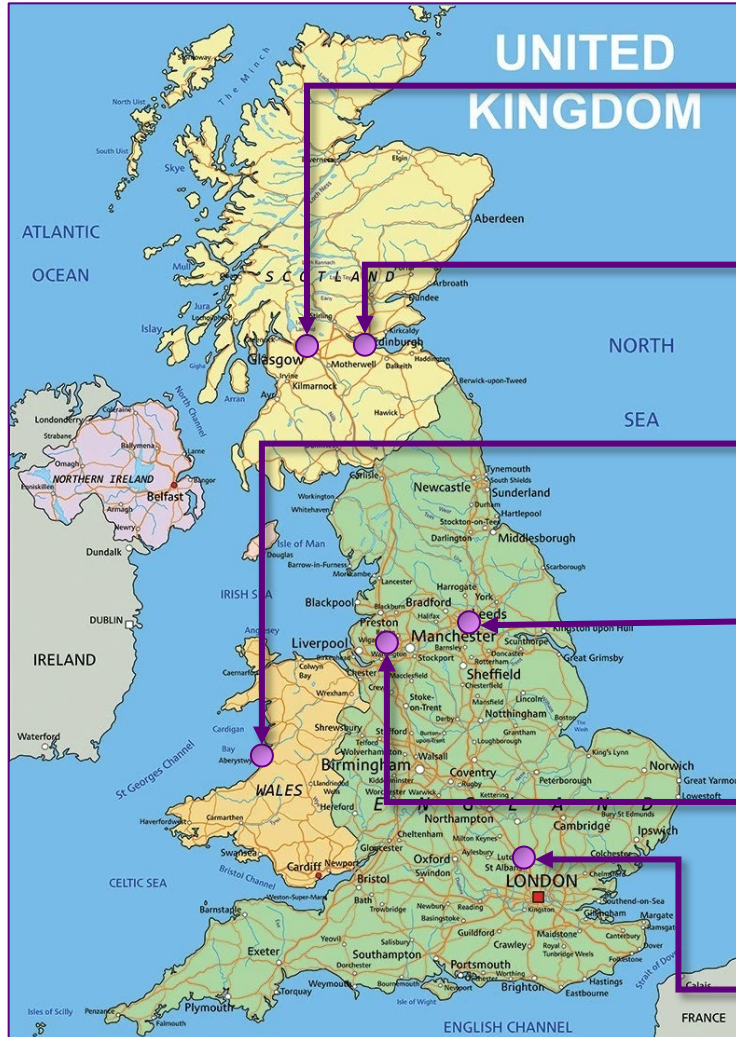
GScan– Corrosion Identification



Atkins Half-joint simulation with section loss defects



Muon Tomography Deployment



M8 Kingston Bridge – Half Joint Investigation



Forth Road Bridge – Main Cable Scan



Llanrhystud Bridge, Wales – Main Deck Scan



4 x Structures North East Region – Half Joint and Post Tensioned Bridges



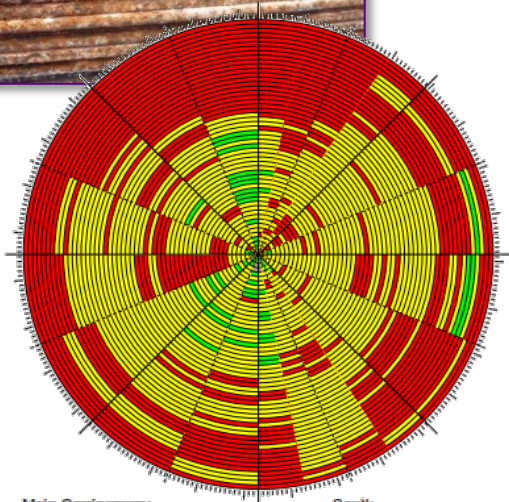
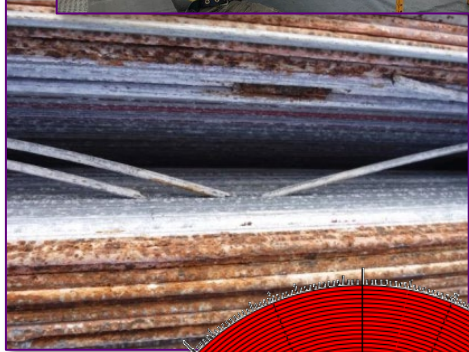
Broughton Circle, North West – Half Joint Bridge



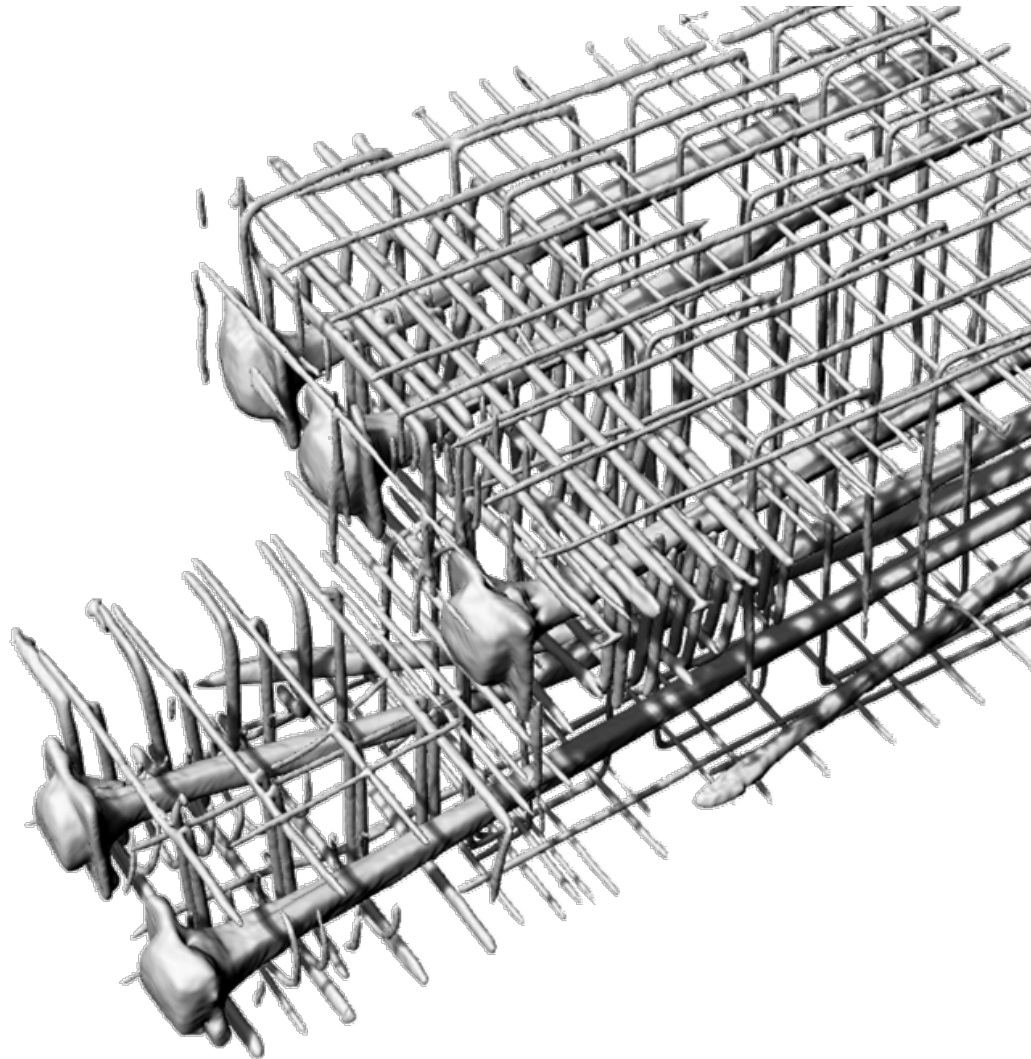
A14 Huntindon Viaduct Section Trials – Half Joints and Post Tensioned Sections



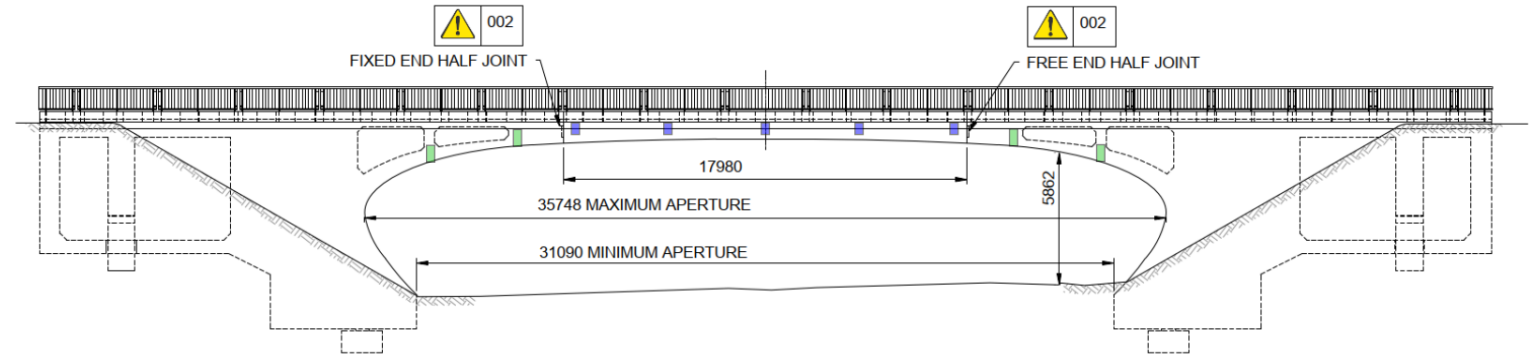
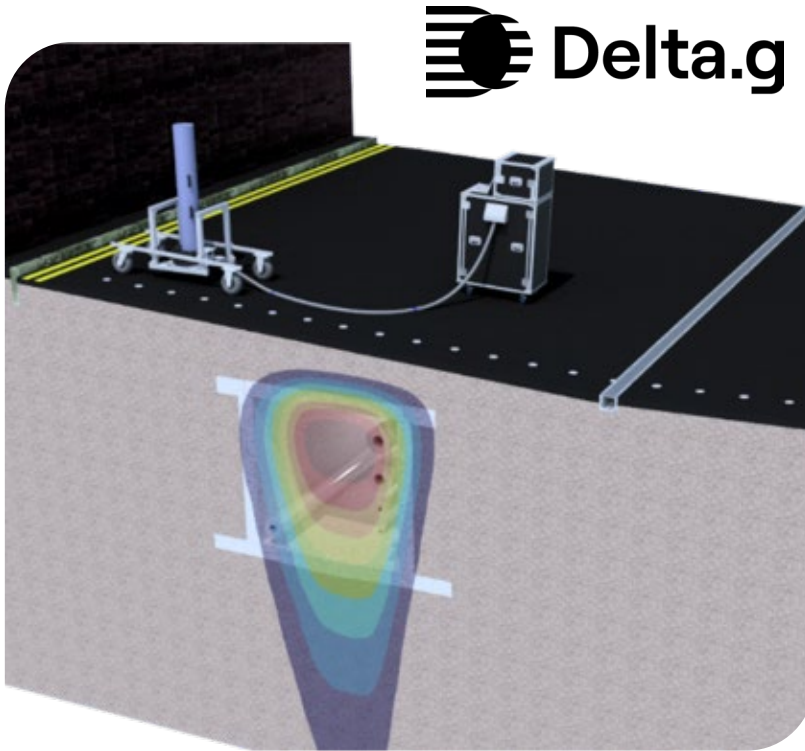
MST for Suspension Bridge Main Cables



Augmented Reality Trials



Quantum Technologies



Target Outcomes

Void detection within bridge structure

Identification of water within voids

Determination of concrete extents (including piles)

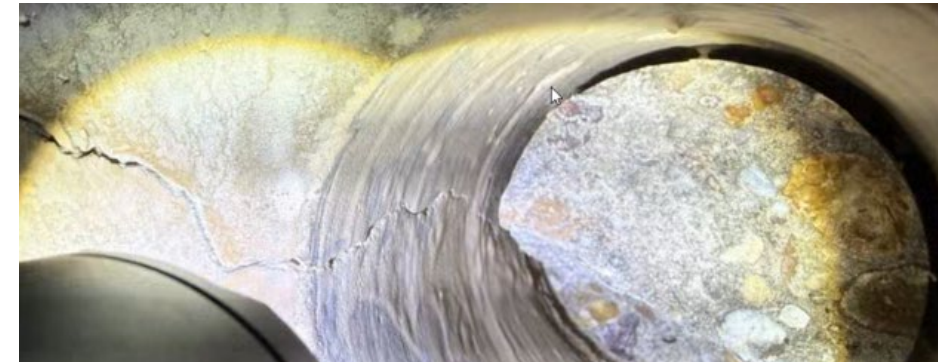
Quantum Gravity Gradiometry

2nd National Highways Structures Moonshot **Conference** – February 2027

Quantum Technologies



Quantum Endoscopy



2nd National Highways Structures Moonshot **Conference** – February 2027

Thank You

